

EPC2034C – Enhancement Mode Power Transistor

 $V_{DS}, 200\text{ V}$
 $R_{DS(on)}, 8\text{ m}\Omega$
 $I_D, 48\text{ A}$


Halogen-Free

Gallium Nitride's exceptionally high electron mobility and low temperature coefficient allows very low $R_{DS(on)}$, while its lateral device structure and majority carrier diode provide exceptionally low Q_G and zero Q_{RR} . The end result is a device that can handle tasks where very high switching frequency, and low on-time are beneficial as well as those where on-state losses dominate.

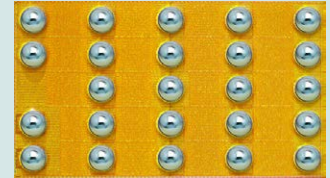
Maximum Ratings

PARAMETER		VALUE	UNIT
V_{DS}	Drain-to-Source Voltage (Continuous)	200	V
I_D	Continuous ($T_A = 25^\circ\text{C}$)	48	A
	Pulsed (25°C , $T_{PULSE} = 300\text{ }\mu\text{s}$)	213	
V_{GS}	Gate-to-Source Voltage	6	V
	Gate-to-Source Voltage	-4	
T_J	Operating Temperature	-40 to 150	$^\circ\text{C}$
T_{STG}	Storage Temperature	-40 to 150	

Thermal Characteristics

PARAMETER		TYP	UNIT
$R_{\theta JC}$	Thermal Resistance, Junction-to-Case	0.3	$^\circ\text{C}/\text{W}$
$R_{\theta JB}$	Thermal Resistance, Junction-to-Board	4	
$R_{\theta JA}$	Thermal Resistance, Junction-to-Ambient (Note 1)	45	

Note 1: $R_{\theta JA}$ is determined with the device mounted on one square inch of copper pad, single layer 2 oz copper on FR4 board. See https://epc-co.com/epc/documents/product-training/Appnote_Thermal_Performance_of_eGaN_FETs.pdf for details



EPC2034C eGaN® FETs are supplied only in passivated die form with solder bumps. Die Size: 4.6 mm x 2.6 mm

- High Frequency DC/DC Conversion
- Multi-level AC/DC Power Supplies
- Wireless Power
- Solar Micro Inverters
- Robotics
- Class-D Audio
- Low Inductance Motor Drives

Static Characteristics ($T_J = 25^\circ\text{C}$ unless otherwise stated)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
BV_{DSS}	Drain-to-Source Voltage	$V_{GS} = 0\text{ V}$, $I_D = 0.6\text{ mA}$	200			V
I_{DSS}	Drain-Source Leakage	$V_{DS} = 160\text{ V}$, $V_{GS} = 0\text{ V}$, $T_J = 25^\circ\text{C}$		0.03	0.4	mA
I_{GSS}	Gate-to-Source Forward Leakage	$V_{GS} = 5\text{ V}$, $T_J = 25^\circ\text{C}$		0.002	4	mA
	Gate-to-Source Forward Leakage [#]	$V_{GS} = 5\text{ V}$, $T_J = 125^\circ\text{C}$		0.03	9	mA
	Gate-to-Source Reverse Leakage	$V_{GS} = -4\text{ V}$, $T_J = 25^\circ\text{C}$		0.03	0.4	mA
$V_{GS(TH)}$	Gate Threshold Voltage	$V_{DS} = V_{GS}$, $I_D = 7\text{ mA}$	0.8	1.1	2.5	V
$R_{DS(on)}$	Drain-Source On Resistance	$V_{GS} = 5\text{ V}$, $I_D = 20\text{ A}$		6	8	m Ω
V_{SD}	Source-Drain Forward Voltage	$I_S = 0.5\text{ A}$, $V_{GS} = 0\text{ V}$		1.7		V

[#] Defined by design. Not subject to production test.

Dynamic Characteristics ($T_j = 25^\circ\text{C}$ unless otherwise stated)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
C_{ISS}	Input Capacitance [#]	$V_{DS} = 100\text{ V}, V_{GS} = 0\text{ V}$		1155	1386	pF
C_{RSS}	Reverse Transfer Capacitance			3.1		
C_{OSS}	Output Capacitance [#]			641	962	
$C_{OSS(ER)}$	Effective Output Capacitance, Energy Related (Note 2)	$V_{DS} = 0\text{ to }100\text{ V}, V_{GS} = 0\text{ V}$		755		
$C_{OSS(TR)}$	Effective Output Capacitance, Time Related (Note 3)			969		
R_G	Gate Resistance			0.5		Ω
Q_G	Total Gate Charge [#]	$V_{DS} = 100\text{ V}, V_{GS} = 5\text{ V}, I_D = 20\text{ A}$		11.1	13.8	nC
Q_{GS}	Gate to Source Charge	$V_{DS} = 100\text{ V}, I_D = 20\text{ A}$		3.8		
Q_{GD}	Gate to Drain Charge			2.0		
$Q_{G(TH)}$	Gate Charge at Threshold			2.1		
Q_{OSS}	Output Charge [#]	$V_{DS} = 100\text{ V}, V_{GS} = 0\text{ V}$		96	144	
Q_{RR}	Source-Drain Recovery Charge			0		

[#] Defined by design. Not subject to production test.

Note 2: $C_{OSS(ER)}$ is a fixed capacitance that gives the same stored energy as C_{OSS} while V_{DS} is rising from 0 to 50% BV_{DSS} .

Note 3: $C_{OSS(TR)}$ is a fixed capacitance that gives the same charging time as C_{OSS} while V_{DS} is rising from 0 to 50% BV_{DSS} .

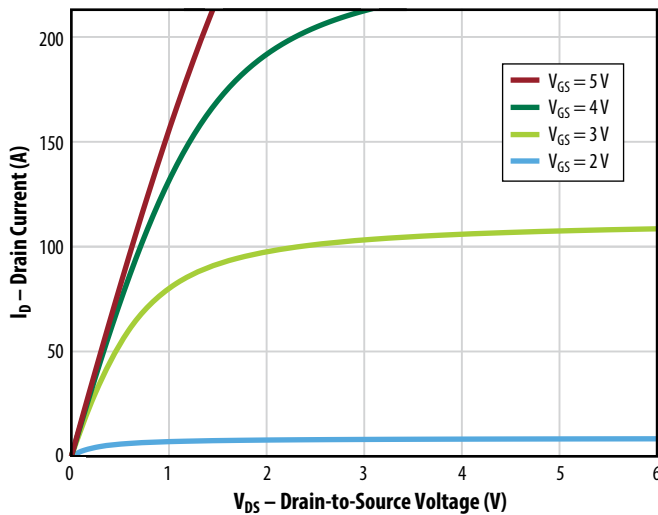
Figure 1: Typical Output Characteristics at 25°C 

Figure 2: Transfer Characteristics

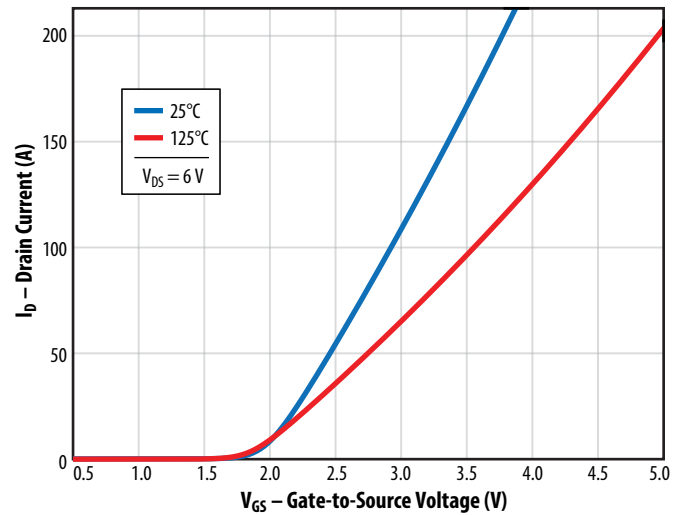
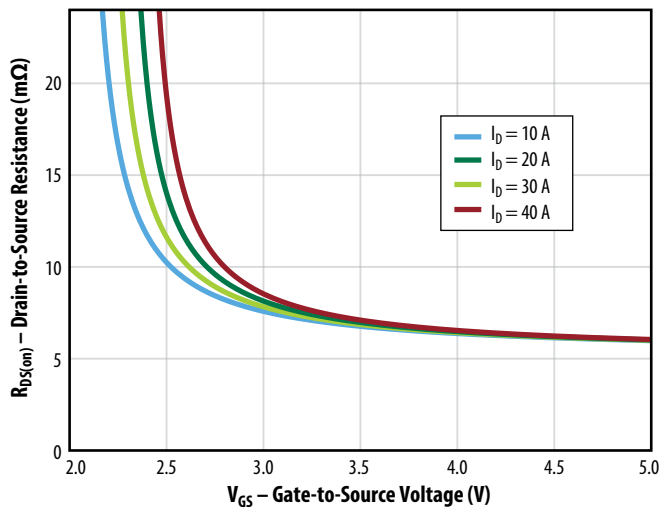
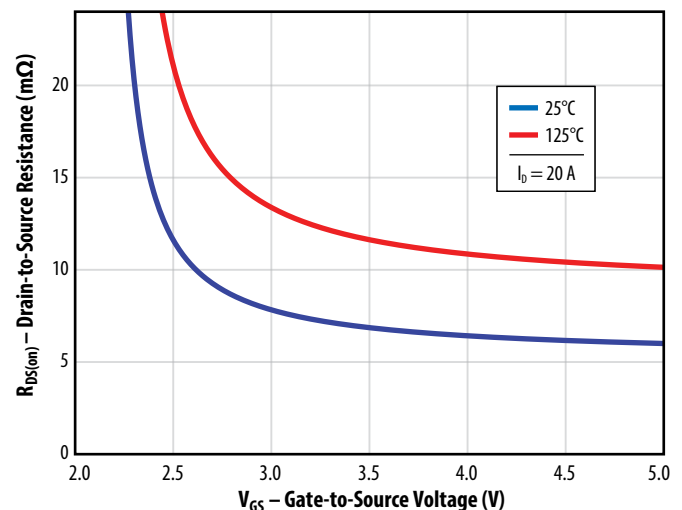
Figure 3: $R_{DS(on)}$ vs. V_{GS} for Various Drain CurrentsFigure 4: $R_{DS(on)}$ vs. V_{GS} for Various Temperatures

Figure 5a: Capacitance (Linear Scale)

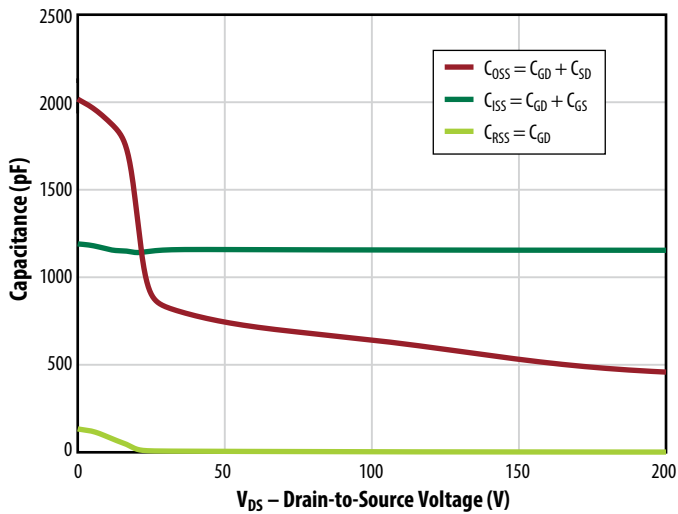


Figure 5b: Capacitance (Log Scale)

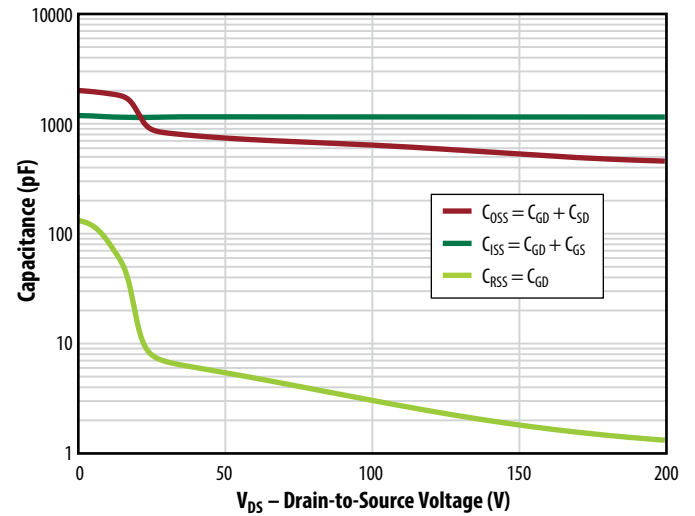
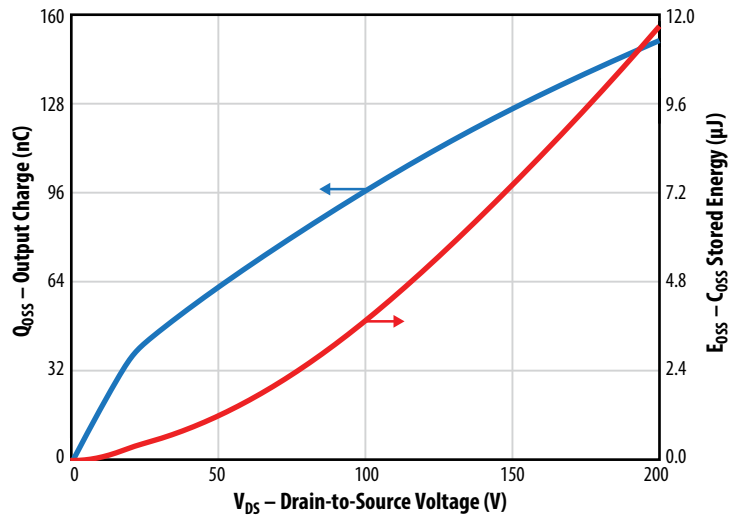
Figure 6: Output Charge and C_{OSS} Stored Energy

Figure 7: Gate Charge

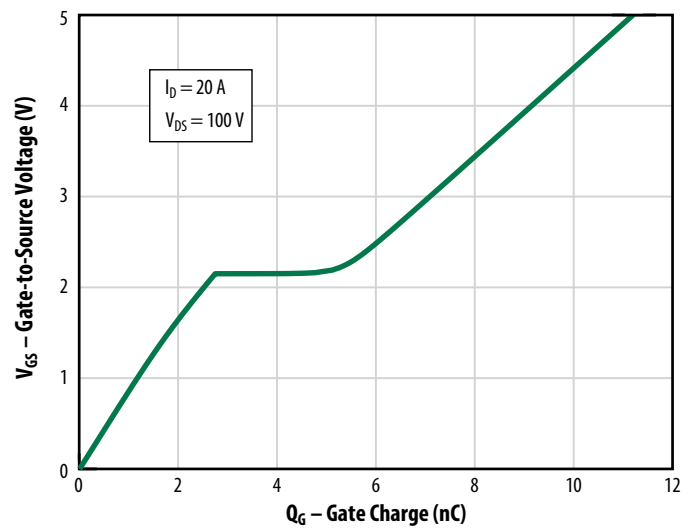


Figure 8: Reverse Drain-Source Characteristics

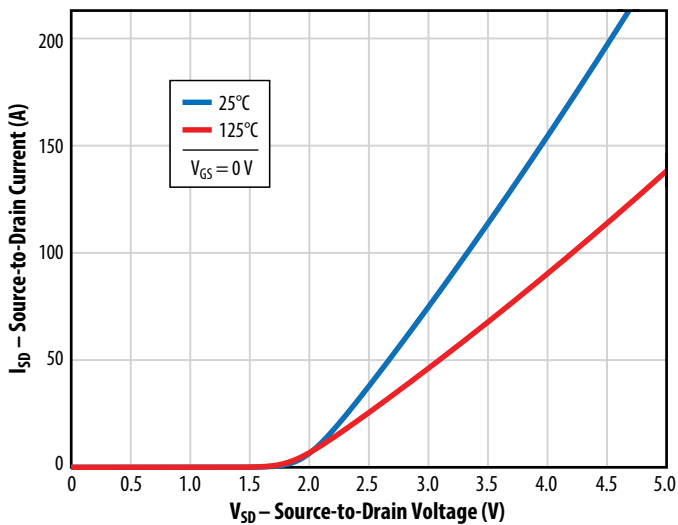
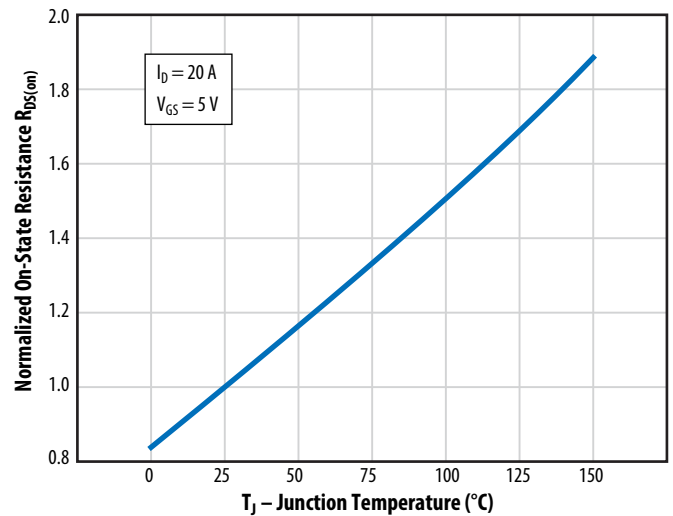


Figure 9: Normalized On-State Resistance vs. Temperature



All measurements were done with substrate shorted to source. $T_J = 25^\circ\text{C}$ unless otherwise stated.

Figure 10: Normalized Threshold Voltage vs. Temperature

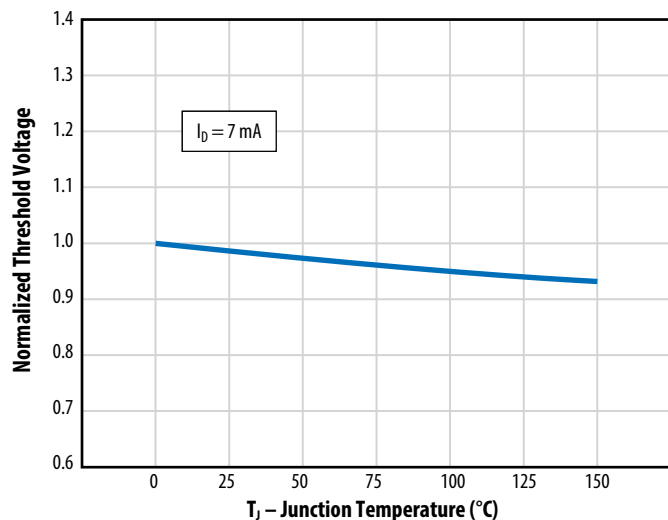


Figure 11: Safe Operating Area

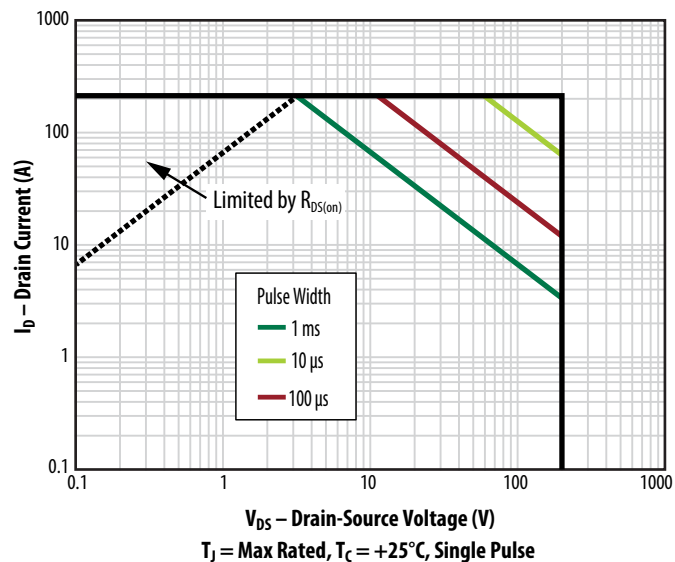
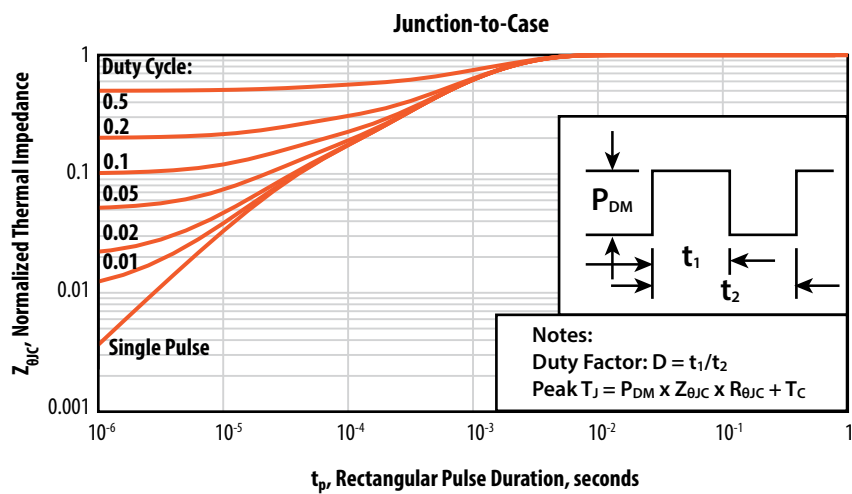
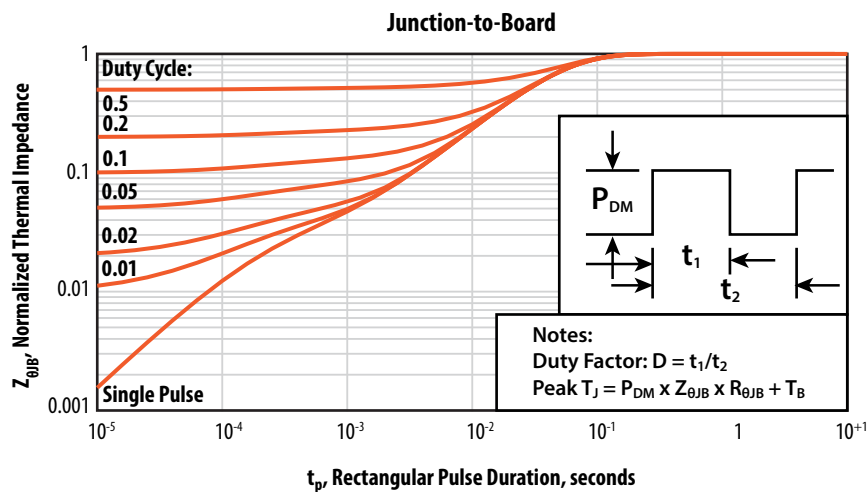
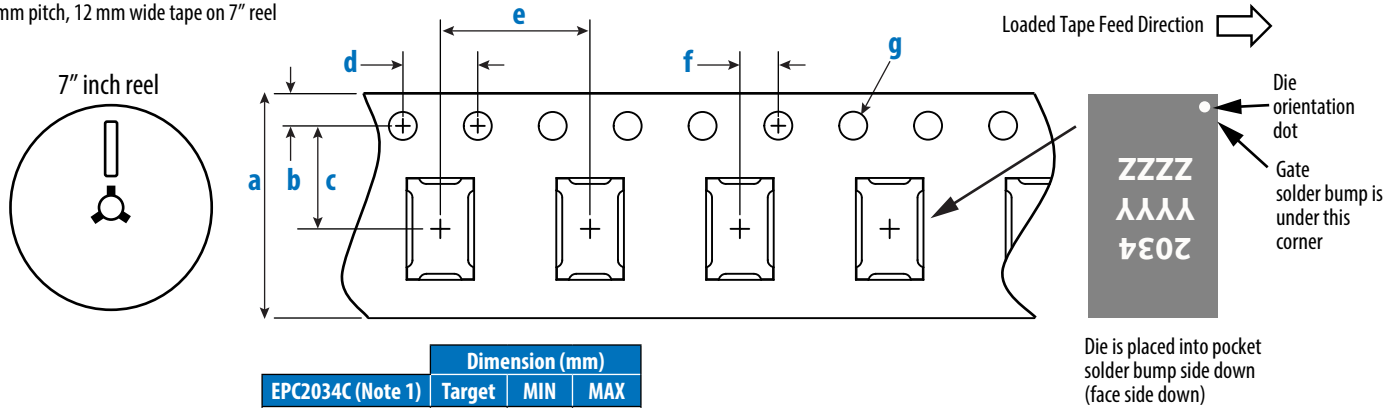


Figure 12: Transient Thermal Response Curves



TAPE AND REEL CONFIGURATION

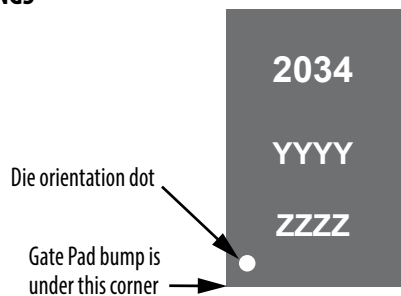
8 mm pitch, 12 mm wide tape on 7" reel



Note 1: MSL 1 (moisture sensitivity level 1) classified according to IPC/JEDEC industry standard.

Note 2: Pocket position is relative to the sprocket hole measured as true position of the pocket, not the pocket hole.

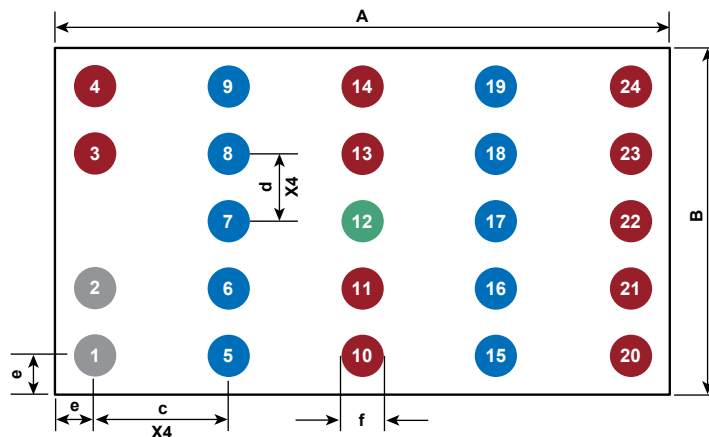
DIE MARKINGS



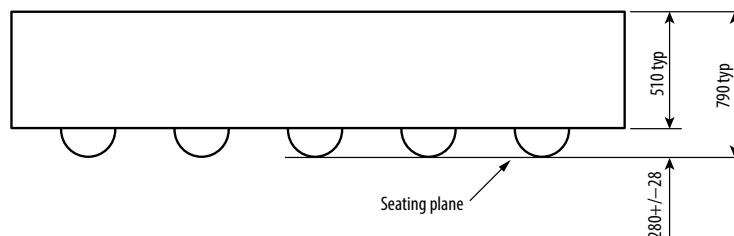
Part Number	Laser Markings		
	Part # Marking Line 1	Lot _ Date Code Marking Line 2	Lot _ Date Code Marking Line 3
EPC2034C	2034	YYYY	ZZZZ

DIE OUTLINE

Solder Bump View



Side View



DIM	Micrometers		
	MIN	Nominal	MAX
A	4570	4600	4630
B	2570	2600	2630
c	1000	1000	1000
d	500	500	500
e	285	300	315
f	332	369	406

Pads 1 and 2 are Gate;

Pads 5, 6, 7, 8, 9, 15, 16, 17, 18, 19 are Drain;

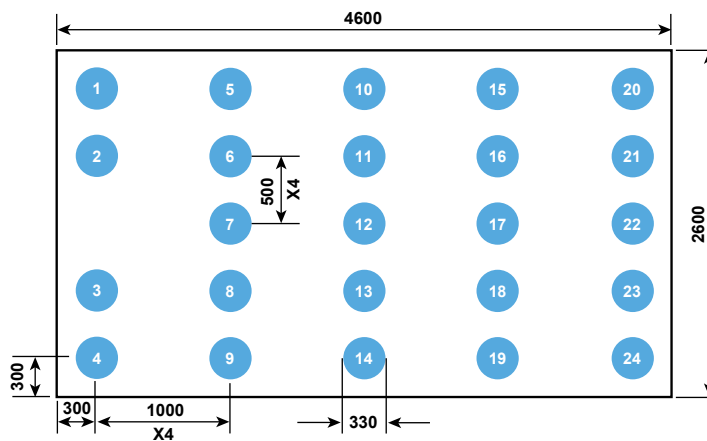
Pads 3, 4, 10, 11, 13, 14, 20, 21, 22, 23, 24 are Source;

Pad 12 is Substrate*

*Substrate pin should be connected to Source

RECOMMENDED LAND PATTERN

(units in μm)



Land pattern is solder mask defined

Solder mask opening is 330 μm

It is recommended to have on-Cu trace PCB vias

Pads 1 and 2 are Gate;

Pads 5, 6, 7, 8, 9, 15, 16, 17, 18, 19 are Drain;

Pads 3, 4, 10, 11, 13, 14, 20, 21, 22, 23, 24 are Source;

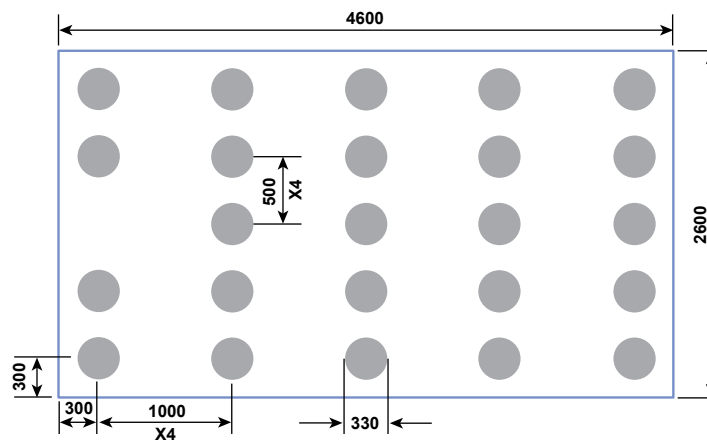
Pad 12 is Substrate*

*Substrate pin should be connected to Source

RECOMMENDED STENCIL DRAWING

(units in μm)

Option 1 : Intended for use with SAC305 Type 4 solder.



Recommended stencil should be 4 mil (100 μm) thick, must be laser cut, openings per drawing.

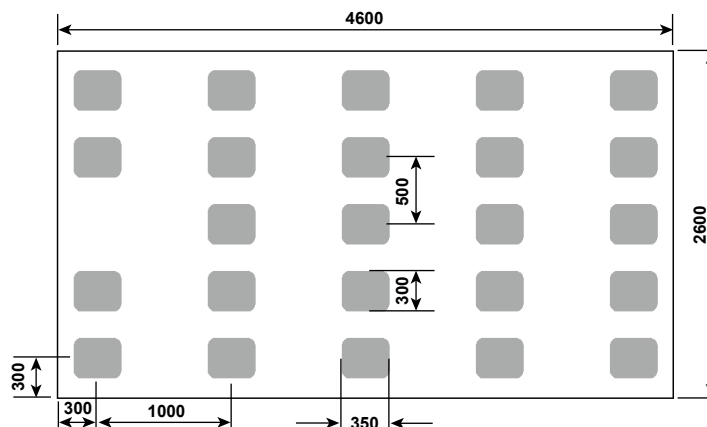
Additional assembly resources available at

<https://epc-co.com/epc/DesignSupport/AssemblyBasics.aspx>

RECOMMENDED STENCIL DRAWING

(units in μm)

Option 2 : Intended for use with SAC305 Type 3 solder.



Recommended stencil should be 4 mil (100 μm) thick, must be laser cut, openings per drawing.

Additional assembly resources available at

<https://epc-co.com/epc/DesignSupport/AssemblyBasics.aspx>

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change without notice.

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